

Eunice Chan

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PUBLICATIONS

^{*}, [†] = equal contribution

Adversarial Bias: Data Poisoning Attacks on Fairness

Eunice Chan, Hanghang Tong

Paper and Presentation at IEEE International Conference on Big Data, 2025

Retrieval-Augmented Zero-Shot Generalization of Logic Rules to Unseen Relations for Temporal Knowledge Graph Forecasting

Eunice Chan, Jiayu Zhou, Lihui Liu, Hanghang Tong

Unpublished manuscript, 2025

On Fairness Vulnerability: A Data Poisoning Attack Perspective

Eunice Chan, Hanghang Tong

Paper and Presentation at Inaugural RISE AI Conference, 2025

AI for Safety and Safety of AI session

Logic Query of Thoughts: Guiding Large Language Models to Answer Complex Logic Queries with Knowledge Graphs

Lihui Liu, Zihao Wang, Ruizhong Qiu, Yikun Ban, **Eunice Chan**, Yangqiu Song, Jingrui He, Hanghang Tong

arXiv pre-print, 2024

Group Fairness via Group Consensus

Eunice Chan^{*}, Zhining Liu^{*}, Yuheng Zhang, Hanghang Tong

Paper and Presentation at 7th Annual ACM FAccT Conference, 2024

Ensuring User-side Fairness in Dynamic Recommender Systems

Hyunsik Yoo, Zhichen Zeng, Jian Kang, Ruizhong Qiu, David Zhou, Zhining Liu, Fei Wang, Charlie Xu, **Eunice Chan**, Hanghang Tong

Paper at ACM The Web Conference, 2024

The Effect of Model Size on Worst-Group Generalization

Alan Le Pham^{*}, **Eunice Chan**^{*}, Vikranth Srivatsa^{*}, Dhruba Ghosh^{*}, Yaoqing Yang, Yaodong Yu, Ruiqi Zhong, Joseph E. Gonzalez[†], Jacob Steinhardt[†]

Paper at NeurIPS Workshop on Distribution Shifts, 2021

Long-Term Motion In-betweening with Motion Graphs

Eunice Chan, Angjoo Kanazawa

Paper and Presentation for University of California, Berkeley Data Science Honors Thesis, 2021

ACADEMIC EXPERIENCE

Student Researcher, *University of Illinois, Urbana-Champaign IDEA Lab (Computer Science)*

Aug 2022 – Dec 2025

- Performed research in incorporating fairness in machine learning. Advised by Professor Hanghang Tong.
- Worked on explainable methods utilizing LLMs over knowledge graphs. Worked with Lihui Liu (lab alumni, current assistant professor at Wayne State University). Advised by Professor Hanghang Tong.

Student Researcher, *University of Illinois, Urbana-Champaign Parasol Lab (Computer Science)*

Aug 2022 – May 2023

- Investigating the utility of living neurons in performing computations as part of the NSF Expedition in Computing project. Focused on applying the paradigm of reservoir computing. Advised by Professor Nancy Amato.
- Performing motion-planning research with a focus on local planners. Advised by Professor Nancy Amato.

SURF Research Fellow, *Stanford University Raymond Lab (Neuroscience)*

Oct 2021 – Nov 2022

- Solder sensors for eye tracking and ran VOR gain up training to study gain consolidation after vestibular stimulation within mouse models under different environmental conditions through the usage of magnetic eye tracking technology.
- Wrote tools to compare and manipulate neural and behavioral signals from SpikeGLX and Spike2 files.
- Worked in the laboratory of Professor Jennifer Raymond with post-doc Trace Stay.

Student Researcher, *University of California, Berkeley RISE Lab (Computer Science)*

Oct 2020 – Apr 2022

- Investigated the effect of a variety of widely-used vision models and augmentation techniques. Designed, planned, and executed experiments. Visualized, and presented results at poster sessions at retreats and weekly meetings.
- Machine learning research focused on model robustness under distributional shift. Researching worst case generalizations with low-count subgroups. Advised by Professor Joseph E. Gonzalez, and Professor Jacob Steinhardt.

Student Researcher, *University of California, Berkeley D-Lab (Computational Social Sciences & Humanities)*

Aug 2019 – Dec 2019

- Machine learning linguistics research focused on exploring efficient speech classification, transcribing audio to IPA representation in Python. Presented at end-of-year symposium.

EDUCATION

University of Illinois, Urbana-Champaign

Aug 2022 – Present

Ph.D. Computer Science

University of California, Berkeley

Aug 2017 – Dec 2021

B.A. Computer Science; B.A. Data Science

Minor in Digital Humanities; Certificate in Entrepreneurship and Technology

TEACHING EXPERIENCE

Teaching Assistant, University of Illinois, Urbana-Champaign Fall 2026

CS 101 Intro Computing: Engineering & Science – Required introductory course for (non-computer science) engineering majors.

Covers basic problem solving, programming techniques, fundamental algorithms, and data structures, with emphasis on applications in the physical sciences and engineering.

Teaching Assistant, University of Illinois, Urbana-Champaign Fall 2025

CS 409 The Art of Web Programming – Upper-level elective on web development.

Covers full-stack web development including the building blocks of the Web (browsers, HTML, CSS, JavaScript) and data exchange (HTTP, AJAX, JSON, REST).

Teaching Assistant, University of Illinois, Urbana-Champaign Fall 2024

CS 410 Text Information Systems – Upper-level elective on information retrieval.

Covers on indexing, ranking, relevance feedback, and evaluation metrics; includes modern search engine architectures and natural language processing components.

Teaching Assistant, University of Illinois, Urbana-Champaign Fall 2023

CS307 Modeling and Learning in Data Science – Required intermediate course for data science majors.

Covers supervised and unsupervised learning, regression, classification, clustering, and model evaluation.

Course Assistant, University of California, Berkeley Summer 2019

CS70 Discrete Mathematics and Probability Theory – Required introductory course for computer science majors.

Covers proof techniques, logic, combinatorics, graph theory, and probability.

Course Assistant, University of California, Berkeley Spring 2019

CS61A Structure and Interpretation of Computer Programs – Required introductory course for computer science majors.

Covers abstraction, recursion, higher-order functions, and interpreters.

AWARDS

2025 RISE AI Travel Award

2024 FAccT Travel Award

2024 NSF GRFP Honorable Mention

2022 Kleiner Perkins Fellowship Finalist

INDUSTRY EXPERIENCE

Research Scientist Intern, Meta (Generative AI)

June 2025 – Aug 2025

– Worked on using Bayesian optimization to optimize datamix for finetuning Llama models with Hakan Inan.

Software Developer, Herald (fka RunLLM, Aqueduct)

Feb 2022 – May 2025

– Worked at a startup performing Q&A using RAG with LLMs. Worked on all stages of the product, touching data ingestion of text such as documentation, chats such as Slack and Discord, JIRA tickets, retrieval, re-ranking, citation, and surfacing results in chat interfaces on API documentation webpages, ticketing software, and instant messaging platforms such as Slack and Discord.

– Startup founded by Professor Joseph Gonzalez and alumni of the RISE Lab at UC Berkeley based on research done in his lab. Additional advising provided by board member Professor Joseph Hellerstein.

– Used by MotherDuck, anyscale, Databricks, and more.

Co-Founder, CTO, Korion Health

Oct 2021 – Dec 2022

– Prototyped and led development of a stethoscope-based telehealth product with machine-learning-based anomaly detection and preliminary diagnosis solution.

– First place winner of the Pitt Challenge Hackathon 2021 and alumni of the Ascender Incubator and LifeX Accelerator programs.

Team Lead, Software Developer, University of California, Berkeley Division of Data Science

Jan 2021 – Aug 2021

– Led 7-person Infrastructure Team focused on the development of an application tracking system and auto-grading software utilized in courses and programs at UC Berkeley and other universities. Conducted interviewing and hiring, designed and developed features, maintained build infrastructure, and performed code reviews. Set up Github Actions to run unit tests in unittest and integration tests in Selenium for continuous integration.

– Improved operational efficiency via the creation of a custom dashboard summarizing application and project data for administrators. Fullstack development in Django with a PostgreSQL database deployed on Azure App Services.

Founding Backend Engineer, Epiji (Skydeck Startup)

Mar 2020 – Dec 2020

– Applied traditional natural language processing (NLP) techniques on resumes to automatically apply heuristic rules to provide suggestions for strengthening the resume.

– Backend-focused fullstack development utilizing the Ruby on Rails framework, PostgreSQL database, Amazon S3, Google OAuth, Redis, Docker, RESTful APIs, and Heroku.

– Part of the SkyDeck Accelerator HotDesk Fall 2020 cohort (<6% acceptance rate)–recipient of the Dean's Seed Fund Grant, and Jack Larson Fellowship.

Natural Language Processing Intern, Oayaw (Skydeck Startup)

Mar 2020 – May 2020

– Applied traditional natural language processing (NLP) techniques on news articles to extract insights and summaries with Python, Amazon Comprehend, SQL server, and Docker.

LEADERSHIP

President, Treasurer of the *Berkeley Undergraduate Digital Humanities Association*
University of California, Berkeley

Sept 2019 – May 2021

Head of Recruitment of the *Data Bears Student Organization*
University of California, Berkeley

Sept 2018 – Sept 2019